

SOC104 Dynamic Malware Analysis Report: winrar600.exe

Static and Dynamic (Sandbox) Malware Analysis | Verdict: False Positive

Platform	LetsDefend (Malware Playbook, Security Analyst)
Analyst	Asha Latha Panuganti
Verdict	False Positive
Analysis type	Dynamic analysis (sandbox behaviour and network activity via VirusTotal) supported by static analysis (hash, file composition)
Action taken	No containment required; alert closed as benign

1. Summary

The SOC104 rule flagged **winrar600.exe** on host **SusieHost (172.16.17.5)**, and the file was allowed to run. Threat intelligence showed only **1 of 70** antivirus engines detected it. Its bundled and dropped files match a genuine WinRAR 6.00 installer, sandbox network activity showed only normal Microsoft/CDN traffic, and Log Management showed **no outbound connections** from the host around the alert time.

Key analytical step: VirusTotal listed several highly detected "parent" files (60+/70). Instead of treating these as evidence against this file, I recognised them as malware samples that *embed* the legitimate WinRAR installer, which is common behaviour and does not make WinRAR itself malicious.

2. Alert Details

Field	Value
Rule	SOC104 - Malware Detected
Event ID	84
Event time	2021-03-21 13:04:51 (+03:00) = 10:04:51 UTC
Severity	Medium
Alert type	Malware
Host	SusieHost
Source IP	172.16.17.5
File name	winrar600.exe
File size	2.95 MB
MD5	c74862e16bcc2b0e02cadb7ab14e3cd6
SHA-256	aff4bb9b15bccff67a112a7857d28d3f2f436e2e42f11be14930fe496269d573
Device action	Allowed
MITRE (rule)	T1204 - User Execution

MAIN CHANNEL		INVESTIGATION CHANNEL	CLOSED ALERTS
SEVERITY	DATE	RULE NAME	
Medium	2021-03-21T13:04:51+03:00	SOC104 - Malware Detected	
EventID :		84	
Event Time :		2021-03-21T13:04:51+03:00	
Rule :		SOC104 - Malware Detected	
Role :		Security Analyst	
Alert Type :		Malware	
Difficulty :		Easy	
MITRE ATT&CK :		T1204	
File Hash :		c74862e16bcc2b0e02cadb7ab14e3cd6	
File Name :		winrar600.exe	
File Size :		2.95 Mb	
Device Action :		Allowed	
Source Address :		172.16.17.5	
Source Hostname :		SusieHost	

Figure 1: SOC104 alert details

3. Investigation Steps

3.1 Quarantine check

Device Action = Allowed, so the file was not blocked or removed. **Result: not quarantined**. The next step was to determine whether the file was actually malicious.

3.2 File hash analysis (VirusTotal)

- **Detection:** 1/70 (SecureAge only). The other 69 engines, including major vendors, reported it as clean.
- A single detection from one engine, with no specific malware family named, is a common false-positive pattern.

1 / 70
Community Score

1/70 security vendor flagged this file as malicious

Reanalyze Similar More

aff4bb9b15bccf67a112a7857d28d3f2f436e2e42f11be14930fe496269d573

WinRAR.exe

Size: 2.95 MB | Last Analysis Date: 6 days ago

peexe invalid-signature direct-cpu-clock-access signed runtime-modules detect-debug-environment checks-user-input overlay idle via-tor

DETECTION DETAILS RELATIONS BEHAVIOR COMMUNITY 30+

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Security vendors' analysis

SecureAge	Malicious	Acronis (Static ML)	Undetected
AhnLab-V3	Undetected	Alibaba	Undetected
AliCloud	Undetected	ALYac	Undetected
Antiy-AVL	Undetected	Arcabit	Undetected
Arctic Wolf	Undetected	Avast	Undetected
AVG	Undetected	Avira (no cloud)	Undetected

Do you want to automate checks?

Figure 2: VirusTotal detection (1/70)

3.3 Static analysis: file composition

Bundled files are the standard WinRAR components: Rar.exe, UnRAR.exe, rarext.dll, rarext64.dll, Uninstall.exe, WhatsNew.txt and SFX modules. **Dropped files** are only Start Menu shortcuts (WinRAR.lnk, WinRAR help.lnk, Console RAR manual.lnk), which is normal installer behaviour.

PE Resource Parents (5) ⓘ			
Scanned	Detections	Type	Name
2021-06-07	59 / 70	Win32 EXE	e4a9d6f54964d1153949a54c585e1572.virus
2023-01-03	63 / 72	Win32 EXE	3705d2aa7c5a181b4647e7b79d763d7a.virus
2022-10-10	55 / 72	Win32 EXE	25447338805ddbdcbacfc99176211fb.virus
2025-09-29	60 / 69	Win32 EXE	ee5e101fd0cdc7c29251b0f8c34c9591.virus
2024-11-18	64 / 73	Win32 EXE	f29eeb8c8999ab5f0837dfc2124e7d2d7893e...

Bundled Files (19) ⓘ			
Scanned	Detections	File type	Name
2026-04-06	1 / 72	Win32 EXE	Default.SFX
2025-11-27	1 / 72	Win32 EXE	Rar.exe
2026-08-02	0 / 69	Win32 DLL	rarext64.dll
2025-10-18	0 / 72	Win32 DLL	rarext.dll
2026-04-17	0 / 71	Win32 EXE	WinCon.SFX
2026-05-14	0 / 70	Win32 EXE	UnRAR.exe
2025-10-12	0 / 72	Win32 EXE	Zip.SFX
2026-01-18	0 / 69	Win32 EXE	wRAR.exe
2024-07-22	0 / 72	Win32 EXE	Uninstall.exe
2026-01-18	0 / 61	Text	WhatsNew.txt

Figure 3: Bundled files matching WinRAR components

Dropped Files (1.7 K) ⓘ			
Scanned	Detections	File type	Name
2023-09-25	0 / 60	Windows shortcut	WinRAR help.lnk
2024-12-20	0 / 62	Windows shortcut	WinRAR.lnk
2022-07-13	0 / 59	Windows shortcut	Console RAR manual.lnk
2025-01-13	0 / 60	Windows shortcut	WinRAR.lnk
2023-12-29	2 / 61	Windows shortcut	WinRAR.lnk
2022-05-07	0 / 59	Windows shortcut	Console RAR manual.lnk
2024-11-09	0 / 63	Windows shortcut	Console RAR manual.lnk
2024-04-03	0 / 59	Windows shortcut	What is new in the latest version.lnk
2024-09-18	0 / 64	Windows shortcut	Console RAR manual.lnk
2024-12-11	0 / 63	Windows shortcut	Console RAR manual.lnk

Figure 4: Dropped files (shortcuts only)

3.4 Dynamic analysis: sandbox network activity

When the file was executed in a sandbox, the contacted domains and IPs were Microsoft, MSN, Akamai and Fastly infrastructure with 0 detections: normal Windows background traffic, with **no C2 indicators**.

Contacted Domains (43) ⓘ			
Domain	Detections	Created	Registrar
198.187.3.20.in-addr.arpa	0 / 89	-	-
a1666.dscr.akamai.net	0 / 89	1999-03-03	MarkMonitor Inc.
a1672.dscr.akamai.net	0 / 89	1999-03-03	MarkMonitor Inc.
a1858.dscr.akamai.net	0 / 89	1999-03-03	MarkMonitor Inc.
a233.dscr.akamai.net	0 / 89	1999-03-03	MarkMonitor Inc.
api.msn.com	0 / 89	1994-11-10	MarkMonitor Inc.
assets.msn.com	0 / 89	1994-11-10	MarkMonitor Inc.
assets.msn.com-ion.edgesui...	0 / 89	2001-04-02	MarkMonitor Inc.
ax-0001.ax-msedge.net	0 / 89	2024-03-06	-
bg.microsoft.map.fastly.net	0 / 89	-	-

Contacted IP addresses (211) ⓘ			
IP	Detections	Autonomous System	Country
104.112.185.183	0 / 89	20940	US

Figure 5: Contacted domains (Microsoft / CDN infrastructure)

3.5 Interpreting the malicious "parent" files

The Execution Parents and PE Resource Parents lists include files with 55-69 detections. These are **other files that contain or launch** this installer. Malware frequently bundles legitimate utilities such as WinRAR. This makes those parent files malicious, **not** the WinRAR installer itself.

Execution Parents (16) ⓘ			
Scanned	Detections	Type	Name
2026-04-05	69 / 72	Win32 EXE	635fba94e29d0f319b4ab837847de1e8.virus
2024-07-20	0 / 68	ZIP	WinRar-x64-x32.zip
2024-08-10	69 / 74	Win32 EXE	wrar600.exe
2026-08-21	60 / 70	Win32 EXE	42cd3582a84951cf7e8d08c08b2b92ba69eb...
2021-03-04	2 / 59	RAR	75ed1b4ca7053fa1a05a18ac33afe7ed8964f...
2022-09-17	58 / 70	Win32 EXE	WinRAR
2023-02-04	0 / 60	RAR	7de6ece126ab879428aa34d57f0c94b53cb9...
2026-04-27	0 / 66	ZIP	c74862e16bcc2b0e02cadb7ab14e3cd6.zip
2026-07-04	3 / 38	ISO image	WinRAR_OldVer.iso
2022-12-17	0 / 61	RAR	Main Tools.rar

Figure 6: Execution parents: malware embedding WinRAR

3.6 Log correlation

Searched Log Management for 172.16.17.5 around 10:04 UTC on 2021-03-21. **No logs found.** No outbound connections were recorded from the host, which is consistent with a normal software installation.

3.7 Open point: signature tag

VirusTotal tagged the file *invalid-signature* and the community score was -4. Given 69/70 clean results, legitimate installer structure and no malicious network activity, this was not enough to change the verdict. In a production environment I would confirm the Authenticode signer (expected: win.rar GmbH) and check whether the download source was the official site.

4. Verdict and Response

Verdict: False Positive. Legitimate WinRAR 6.00 installer.

- No containment required
- No IOCs added, since adding a legitimate software hash as an IOC risks blocking a business tool
- Alert closed with analyst note

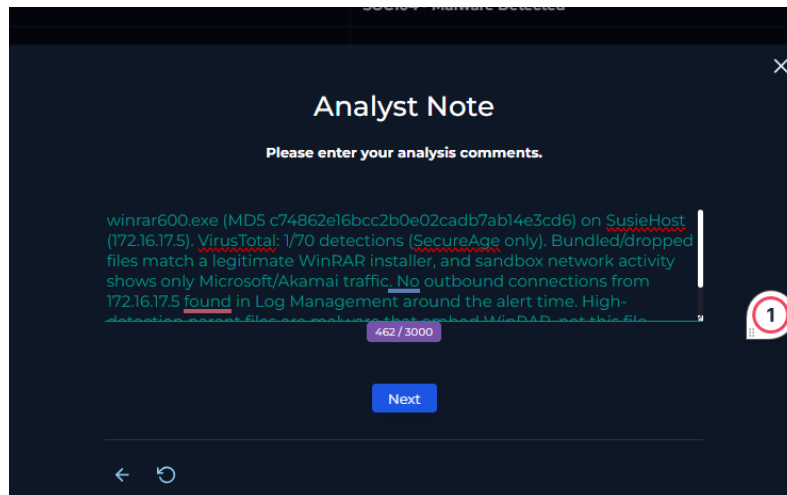


Figure 7: Analyst note submitted

Recommendations:

- Review the SOC104 detection logic to reduce false positives on common signed software
- Consider allow-listing verified WinRAR installer hashes
- Confirm software installation complies with the organisation's software policy

5. Lessons Learned

- **Not every alert is malicious.** Closing false positives accurately is as important as catching true positives, because it keeps analyst time focused on real threats.
- **Weigh the evidence, not a single signal.** One detection, one tag, or one suspicious-looking parent file does not outweigh the overall picture.
- **Parent files are not the file itself.** Legitimate tools are often embedded in malware; context matters.
- **No logs is also evidence.** An absence of outbound connections supports a benign verdict.

Tools used: LetsDefend SIEM, Log Management; VirusTotal (detection, details, relations)